

LAB REPORT: ASSIGNMENT 02

Topic: Creating New Columns Using Python & R

- **Course Name:** Introduction To Data Science
- **Course Code :** ADS 201
- **Course Instructor:** Dr. Muhammad Abul Hasan
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1. Objective

The objective of this assignment is to create new columns in a dataset and apply conditional logic using both Python and R. The assignment demonstrates how to generate calculated values and classify records based on a specified condition.

2. Dataset Overview

- **Dataset Name:** Space_Corrected.csv
- **Description:** The dataset contains historical information about space mission launches, including company names, launch locations, mission details, rocket status, mission status, and other related information. For this assignment, additional practice columns were created to perform calculations and apply conditional operations.

3. Part A: Implementation in Python

Steps Taken:

1. Loaded the dataset using the pandas library.
2. Created two new numeric columns named Score1 and Score2.
3. Calculated the average of the two scores and stored it in a new column named Average_Score.
4. Applied a condition to create a new Status column. If the average score was 85 or above, the status was set to "Pass"; otherwise, it was set to "Fail".
5. Displayed the newly created columns to verify the results..

Python Code:

```

1 import pandas as pd
2
3 # Load dataset
4 df = pd.read_csv("Space_Corrected.csv")
5
6 # 1. Create dummy score columns
7 df["Score1"] = (df.index % 21) + 70      # 70-90
8 df["Score2"] = (df.index % 16) + 75      # 75-90
9
10 # 2. Create 'Average Score' column
11 df["Average Score"] = df[["Score1", "Score2"]].mean(axis=1)
12
13 # 3. Apply condition
14 df["Status"] = df["Average Score"].apply(
15     lambda x: {"_ge_": "Pass" if x >= 85 else "Fail"}
16 )
17
18 # Display output
19 print(df[["Score1", "Score2", "Average Score", "Status"]].head())

```

Output & Observations:

The Python program successfully created four new columns: Score1, Score2, Average_Score, and Status. The average score was calculated correctly, and each record was classified as Pass or Fail based on the specified condition.

Run assignment_2

	Score1	Score2	Average Score	Status
0	70	75	72.5	Fail
1	71	76	73.5	Fail
2	72	77	74.5	Fail
3	73	78	75.5	Fail
4	74	79	76.5	Fail

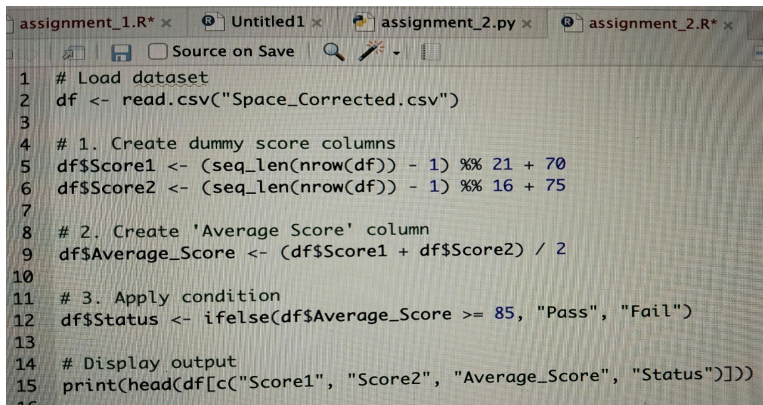
Process finished with exit code 0

4. Part B: Implementation in R

Steps Taken:

1. Loaded the dataset using the read.csv() function.
2. Created two new numeric columns named Score1 and Score2.
3. Calculated the average score and stored it in the Average_Score column.
4. Applied conditional logic using ifelse() to create the Status column.
5. Displayed the generated columns to verify the output.

R Code:



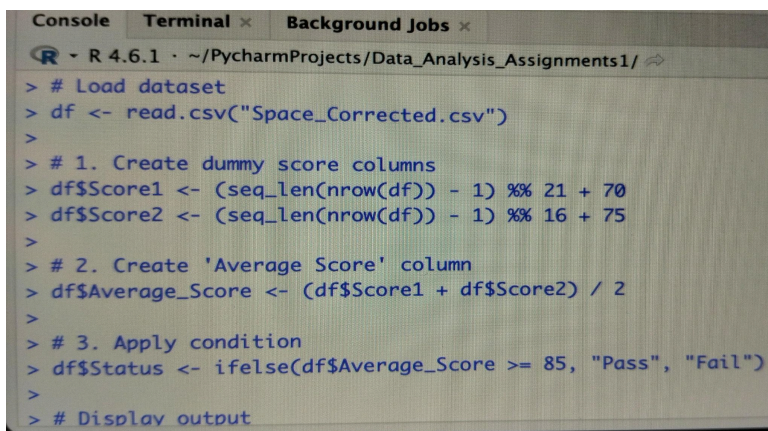
```

1 # Load dataset
2 df <- read.csv("Space_Corrected.csv")
3
4 # 1. Create dummy score columns
5 df$Score1 <- (seq_len(nrow(df)) - 1) %% 21 + 70
6 df$Score2 <- (seq_len(nrow(df)) - 1) %% 16 + 75
7
8 # 2. Create 'Average Score' column
9 df$Average_Score <- (df$Score1 + df$Score2) / 2
10
11 # 3. Apply condition
12 df$Status <- ifelse(df$Average_Score >= 85, "Pass", "Fail")
13
14 # Display output
15 print(head(df[c("Score1", "Score2", "Average_Score", "Status")]))

```

Output & Observations:

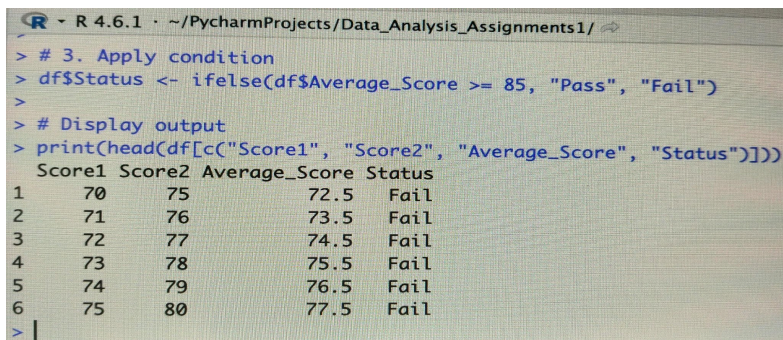
The R program successfully generated the required columns and produced the same results as the Python implementation. The average score was calculated correctly, and the records were classified according to the specified condition.



```

> # Load dataset
> df <- read.csv("Space_Corrected.csv")
>
> # 1. Create dummy score columns
> df$Score1 <- (seq_len(nrow(df)) - 1) %% 21 + 70
> df$Score2 <- (seq_len(nrow(df)) - 1) %% 16 + 75
>
> # 2. Create 'Average Score' column
> df$Average_Score <- (df$Score1 + df$Score2) / 2
>
> # 3. Apply condition
> df$Status <- ifelse(df$Average_Score >= 85, "Pass", "Fail")
>
> # Display output

```



```

> # 3. Apply condition
> df$Status <- ifelse(df$Average_Score >= 85, "Pass", "Fail")
>
> # Display output
> print(head(df[c("Score1", "Score2", "Average_Score", "Status")]))
  Score1 Score2 Average_Score Status
1     70     75          72.5   Fail
2     71     76          73.5   Fail
3     72     77          74.5   Fail
4     73     78          75.5   Fail
5     74     79          76.5   Fail
6     75     80          77.5   Fail
> |

```

5. Conclusion

Both Python and R successfully completed the required tasks for this assignment. New columns were created, the average score was calculated correctly, and conditional logic was applied to classify each record. The results produced by both programming languages were consistent, demonstrating that the same data manipulation task can be performed effectively in either environment.